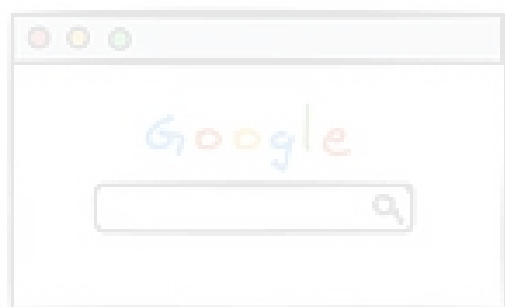


Tool / Retrieval Layer



Web Search



Vector DB



Operational
Data

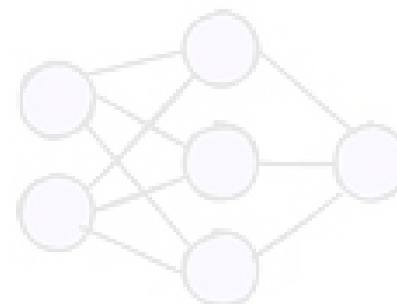


business
logic

Why RAG Architectures Fail at Scale

5 hidden bottlenecks and the engineering fixes that solve them

Reasoning Layer



LLM

learn

Introduction

It worked perfectly with 10K docs during testing.

But two months after launch, you hit real scale:

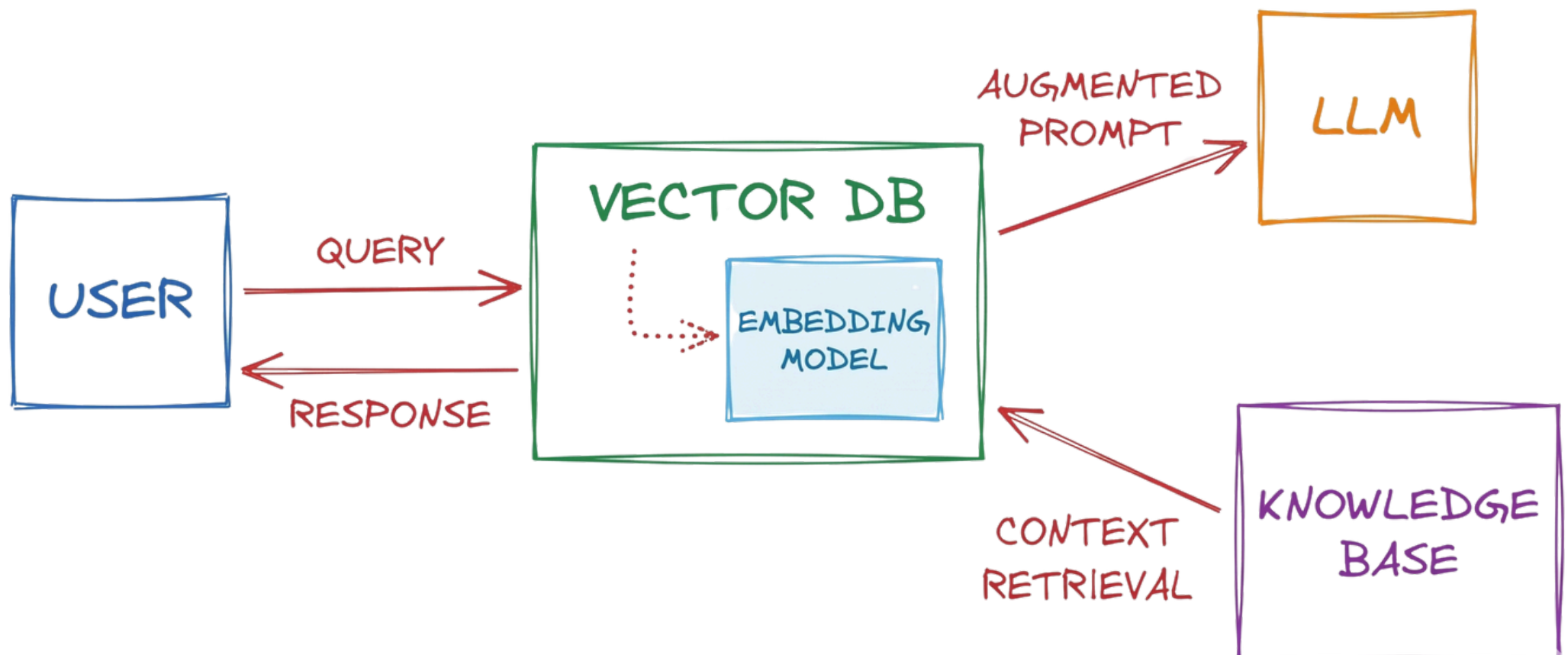
- Latency: **400ms → 3.2 seconds (p95)**
- Costs: **\$2K → \$14K/month**
- Vector DB: **Crashed (twice)**

The simple architecture that survived testing is now failing your users.

This carousel dives into 5 common RAG architecture bottlenecks - and the specific engineering changes required to fix them.

Below is a simple Honeymoon Architecture we will try to keep on applying fixes to this architecture to get our perfect scalable RAG architecture.

THE HONEYMOON ARCHITECTURE



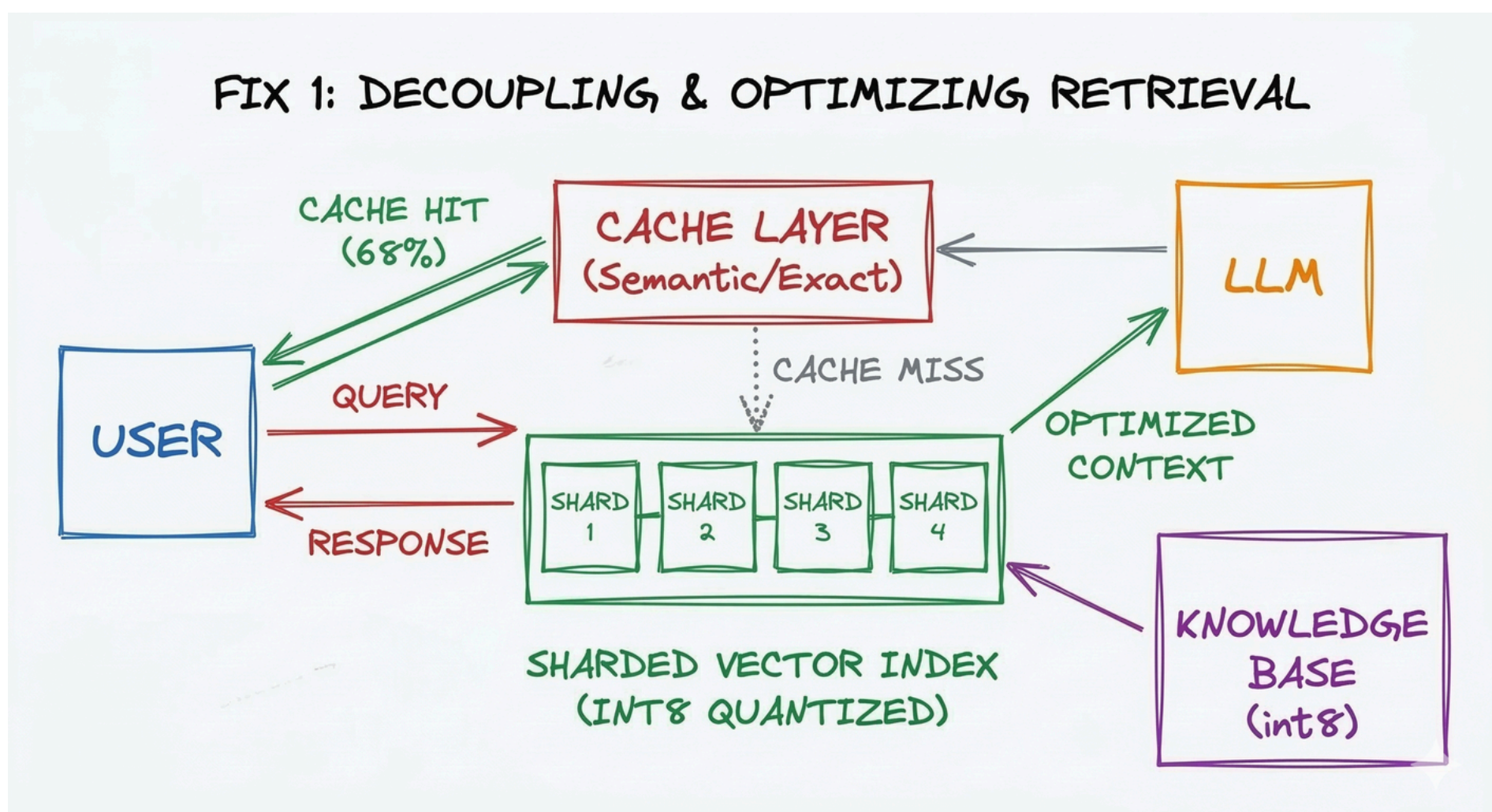
Latency Explosion

The POC worked at 10K docs. The system failed at 50M docs.

You were running a monolithic Vector DB on a single node. At 50M vectors, HNSW index traversal became memory-bound, causing critical memory swapping and disk page faults.

The Fix: A 3-Part Architecture Refactor

1. Semantic Caching: Offloads the DB; 68% of queries never hit the index.
2. Vector Quantization (fp32 $\rightarrow$ int8): Reduces memory footprint by 4x.
3. Horizontal Sharding: Distributes the load across nodes.

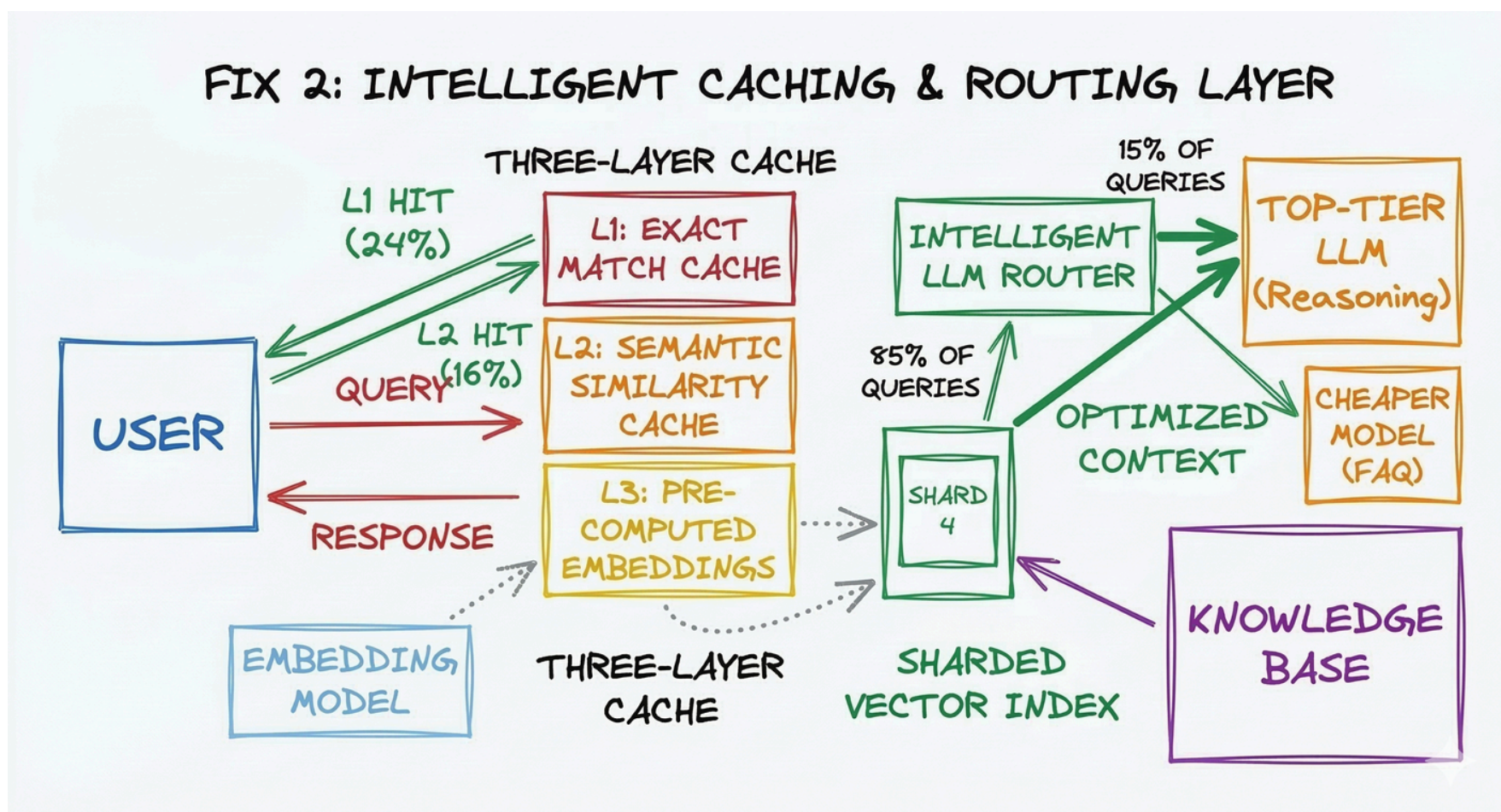


COST SPIRAL

You launched RAG, it worked, people used it. Then accounting pinged you about the AI bill.

A query audit revealed the cause: **56% of your queries were identical duplicates or paraphrases.**

Every single one was triggering a new LLM call (\$0.025/call). You were spending \$5,100/month re-answering the same questions.

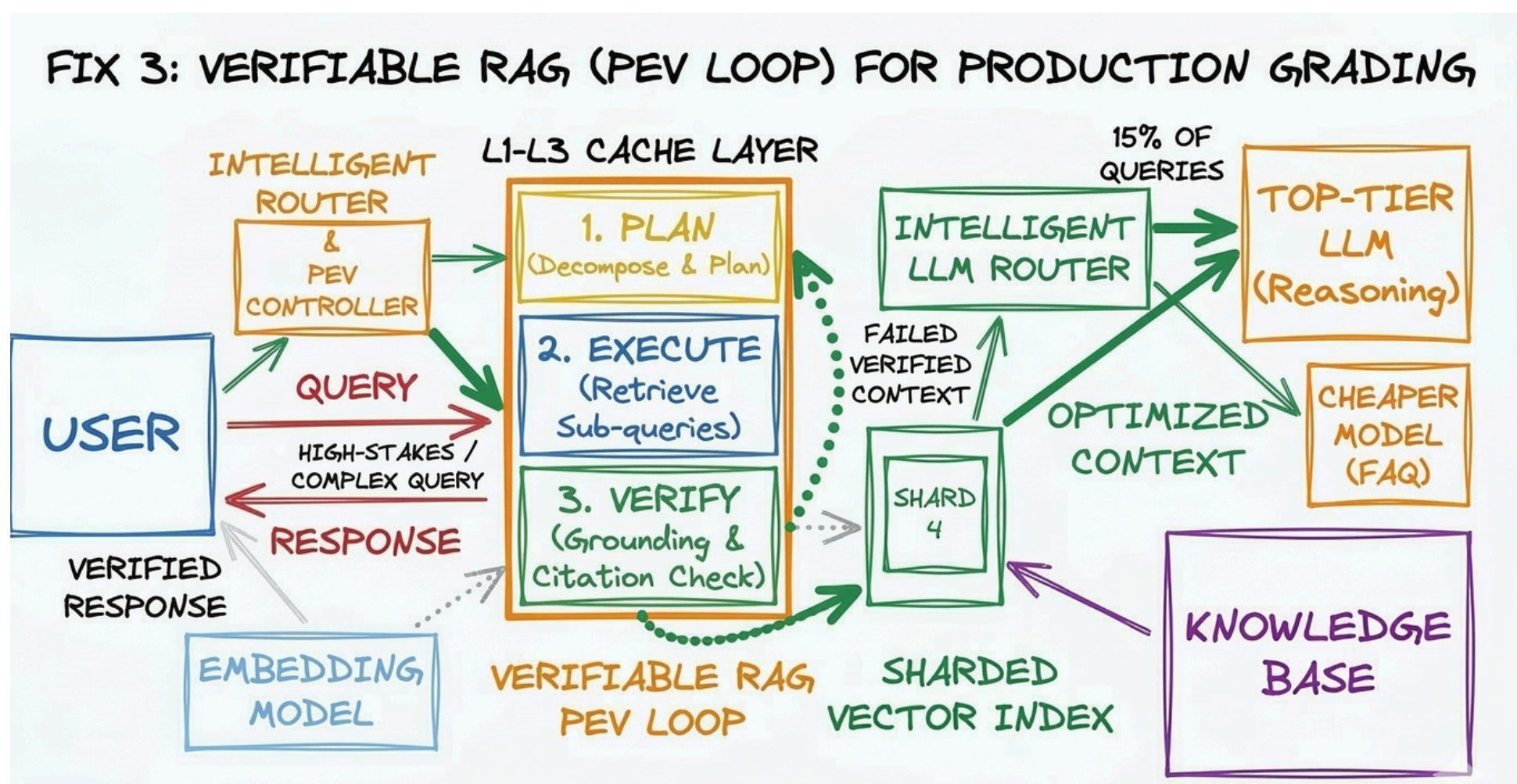


The Fix:

1. L1 Exact Match: (24% hit rate)
2. L2 Semantic Cache: (16% hit rate for paraphrases)
3. L3 Pre-computed Embeddings: (Saves 28% on encoding)
4. Intelligent Query Routing (routing simple queries to cheaper models).

HALLUCINATIONS

At small scale, hallucinations are annoying. At production scale, they are dangerous. **Ex:** An employee asks: "How much PTO can I carry over?" RAG retrieves the 2023 policy (the wrong document). The LLM confidently hallucinations: "10 days maximum." (The actual 2024 policy requires 0 days carryover.)



The Fix: The "Verifiable RAG" (Plan-Execute-Verify) Loop

1. Complexity routing: Verify high-stakes queries
2. Plan–retrieve–reason: Break queries into sub-questions
3. Grounding check: Ensure every claim has a citation

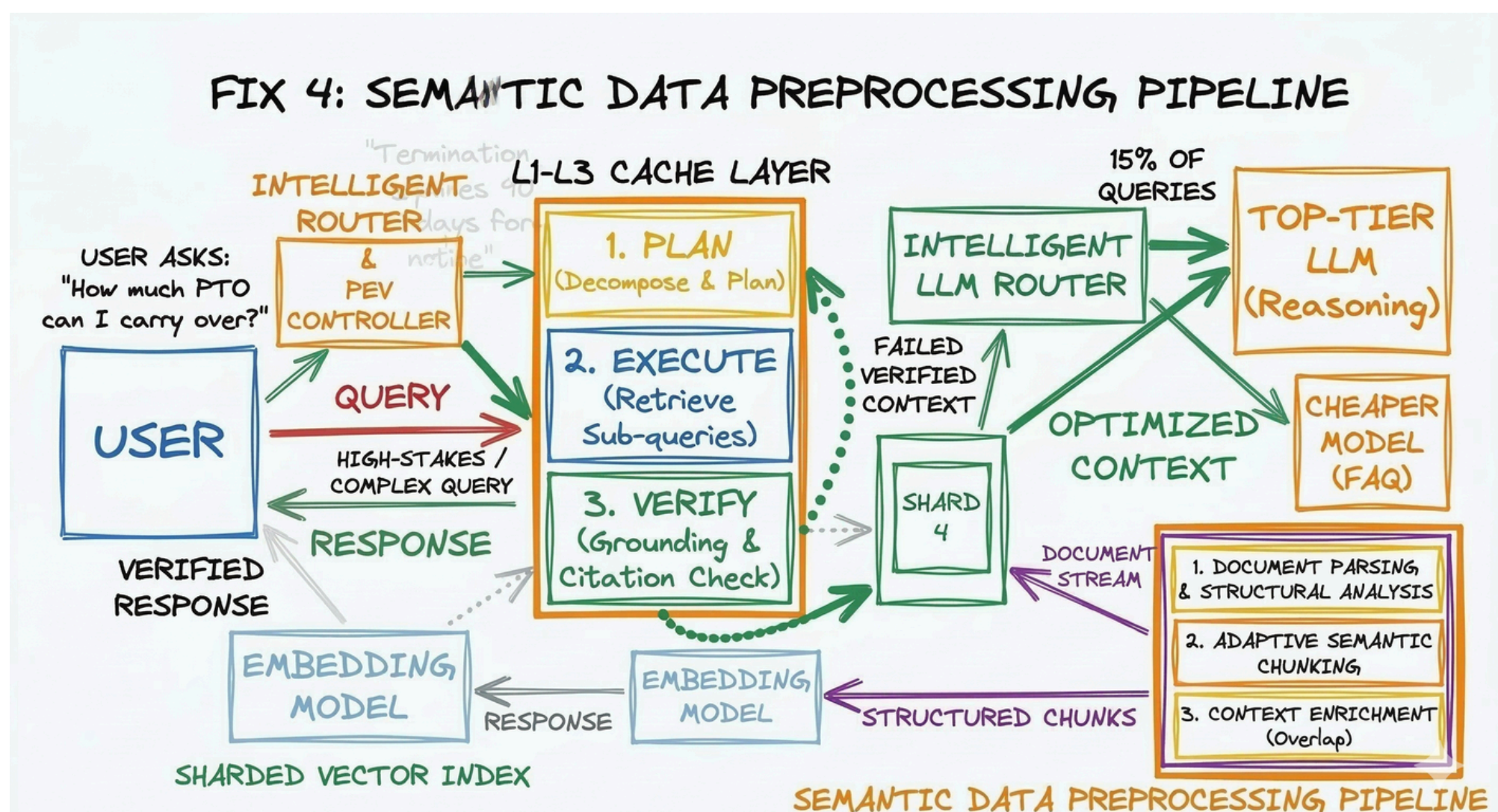
RETRIEVAL ACCURACY COLLAPSES

You processed 24K pages of documentation using 512-token chunks with no overlap.

In production, users reported cut-off code examples, incomplete instructions, and broken tables.

The Fix: Adaptive Semantic Chunking

1. Semantic Similarity over Token Counts: Chunk by meaning, not arbitrary count.
2. Structural Integrity: Keep code blocks, tables, and lists intact.
3. Context Preservation: Add 10-20% strategic overlap.

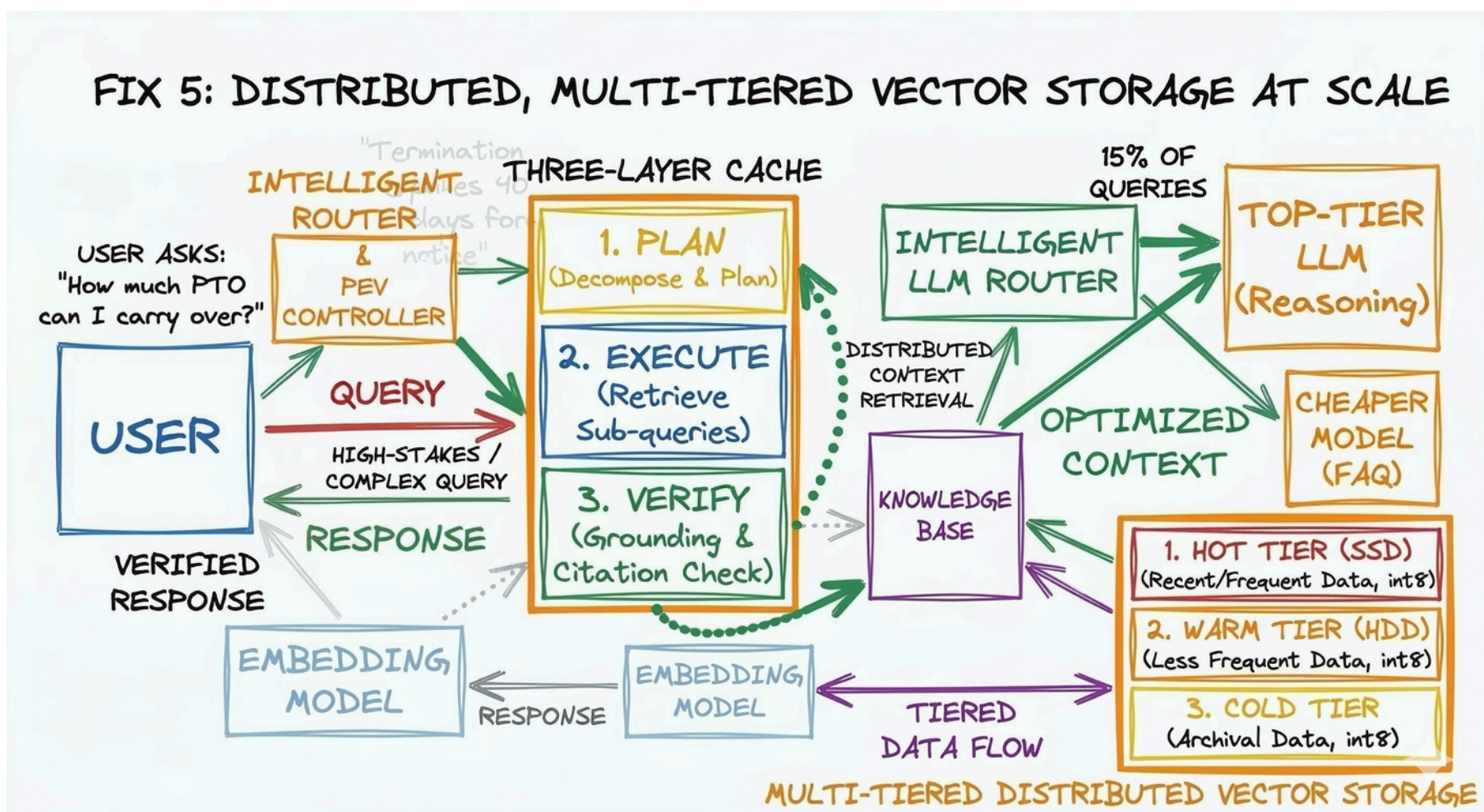


VECTOR DB HITS A WALL

It worked perfectly at 10M vectors (P95 <200ms). Everything broke when you crossed 50M vectors. Vertically scaling didn't help.

The true cost of trying to run a distributed workload on a single-node architecture was revealed:

- P95 Latency: 200ms → 4+ seconds.
- Memory Usage: 98% constant, causing 2-3 OOM crashes per week.
- Index Rebuilds: 4 hours → 28+ hours (it literally took longer than a day).



The Fix: **The Distributed, Multi-Tiered Architecture**

1. Sharding: Split index to parallelize queries
2. Quantization: 4× less memory
3. Tiered storage: SSD for hot data, cheap storage for cold data

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My Neural Network:



Object Detection with Large Vision Language Models (LVLMs)

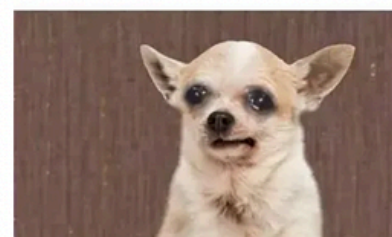
Object detection, now smarter with LVLMs

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